



Configuración de VOIP con Asterisk y Zoiper

Arquitectura

- Servidor Linux con Asterisk
- Cliente Windows con Zoiper
- Cliente Android con Zoiper

Instalación y configuración de Zoiper

Instalamos Asterisk en nuestro servidor con el siguiente comando

```
victor@victor: ~  
victor@victor:~$ sudo apt install asterisk  
[sudo] password for victor:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40  
  linux-image-6.8.0-40-generic linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic  
  linux-tools-6.8.0-40-generic  
Use 'sudo apt autoremove' to remove them.
```

Una vez instalado, con `sudo nano` modificamos los archivos `/etc/asterisk/cel.conf` y `/etc/asterisk/cdr.conf` para incluir las rutas necesarias en caso de que no vengan preconfiguradas

```
[radius]  
;  
; Log date/time in GMT  
;usegmttime=yes  
;  
; Set this to the location of the radiusclient-ng configuration file  
; The default is /etc/radiusclient-ng/radiusclient.conf  
;radiuscfg => /usr/local/etc/radiusclient-ng/radiusclient.conf  
;  
radiuscfg => /etc/radcli/radiusclient.conf
```

```
GNU nano 6.2 /etc/asterisk/cdr.conf  
[radius]  
;usegmttime=yes ; log date/time in GMT  
;loguniqueid=yes ; log uniqueid  
;loguserfield=yes ; log user field  
; Set this to the location of the radiusclient-ng configuration file  
; The default is /etc/radiusclient-ng/radiusclient.conf  
;radiuscfg => /usr/local/etc/radiusclient-ng/radiusclient.conf  
  
radiuscfg => /etc/radcli/radiusclient.conf
```


Reiniciamos el servicio con `sudo service asterisk reload` y `sudo systemctl restart asterisk`

```
victor@victor:~$ sudo service asterisk reload
victor@victor:~$ sudo systemctl restart asterisk
victor@victor:~$ sudo systemctl status asterisk
● asterisk.service - Asterisk PBX
   Loaded: loaded (/lib/systemd/system/asterisk.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-02-05 17:27:13 CET; 8s ago
     Docs: man:asterisk(8)
  Main PID: 47817 (asterisk)
    Tasks: 80 (limit: 2210)
   Memory: 42.4M
      CPU: 681ms
   CGroup: /system.slice/asterisk.service
           └─47817 /usr/sbin/asterisk -g -f -p -U asterisk
             └─47820 astcanary /var/run/asterisk/alt.asterisk.canary.tweet.tweet.tweet 47817

feb 05 17:27:12 victor systemd[1]: Starting Asterisk PBX...
feb 05 17:27:13 victor systemd[1]: Started Asterisk PBX.
victor@victor:~$
```

Una vez reiniciado el servicio, pasamos a configurar el archivo `/etc/asterisk/sip.conf`. Configuramos los puertos y direcciones en la sección `[general]`. La `bind address 0.0.0.0` indica que se escuchara cualquier ip.

```
[general]
context=public                ; Default context for incoming calls. Defaults to 'default'

port=5060
bindaddr=0.0.0.0
disallow=all
allow=alaw
allow=ulaw
qualify=yes
```

Creamos la sección `[user]` que usaremos como 'clase' para replicar la configuración importante para varios usuarios. Les daremos nombre y contraseña

```
GNU nano 6.2 /etc/asterisk/sip.conf *
[user]
type=friend
host=dynamic
context=ser-t12

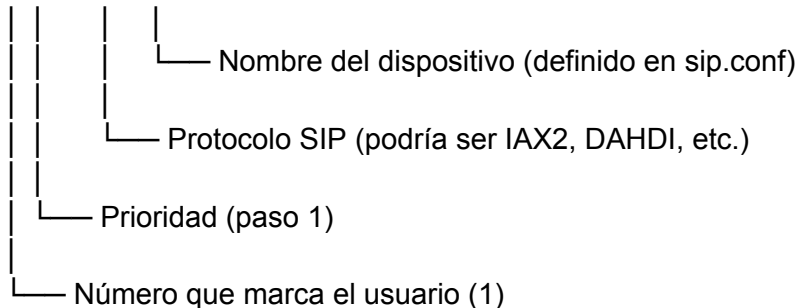
[ext1](user)
username=profesor
secret=1234

[ext2](user)
username=alumno-1
secret=1234

[ext3](user)
username=alumno-2
secret=1234
```

Guardamos el archivo y modificamos `/etc/asterisk/extensions.conf`. Creamos la sección `[ser-t12]` y escribimos nuestras extensiones. Con la palabra clave `exten` le daremos un número de marcado, orden de prioridad, protocolo de llamada y nombre del dispositivo en `sip.conf`

```
exten => 1,1,Dial(SIP/ext1)
```



```
GNU nano 6.2 /etc/asterisk/extensions.conf *
[ser-t12]
exten => 1,1,Dial(SIP/ext1)
exten => 2,1,Dial(SIP/ext2)
exten => 3,1,Dial(SIP/ext3)
```

Guardamos la configuración y recargamos el servicio.

```
victor@victor:~$ sudo service asterisk reload
victor@victor:~$
```

Con el comando `sudo asterisk -rvvv` accedemos a la consola de asterisk

```
victor@victor:~$ sudo asterisk -rvvv
```

Dentro de esta CLI podemos comprobar que se han guardado los usuarios correctamente con `sip show users`

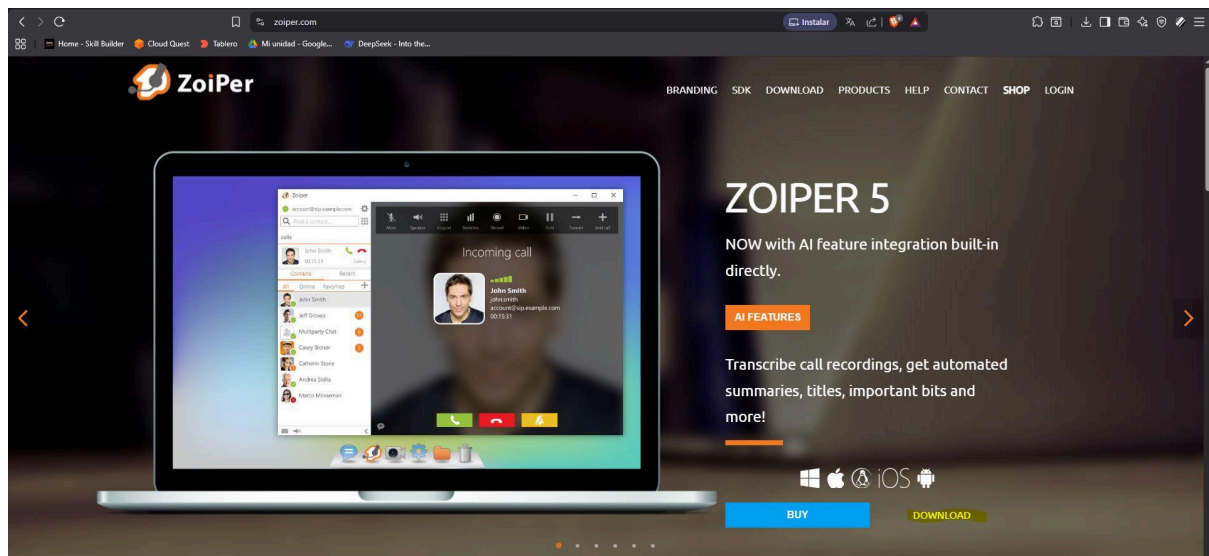
```
victor@victor:~$ sudo asterisk -rvvv
Asterisk 18.10.0~dfsg+~cs6.10.40431411-2, Copyright (C) 1999 - 2021, Sangoma Technologies Corporation
and others.
Created by Mark Spencer <markster@digium.com>
Asterisk comes with ABSOLUTELY NO WARRANTY; type 'core show warranty' for details.
This is free software, with components licensed under the GNU General Public
License version 2 and other licenses; you are welcome to redistribute it under
certain conditions. Type 'core show license' for details.
=====
Connected to Asterisk 18.10.0~dfsg+~cs6.10.40431411-2 currently running on victor (pid = 47817)
victor*CLI> sip show users
Username      Secret      Accountcode  Def.Context  ACL  Forcerport
ext1          1234        1234         ser-t12      No   No
ext2          1234        1234         ser-t12      No   No
ext3          1234        1234         ser-t12      No   No
user          1234        1234         ser-t12      No   No
victor*CLI>
```

Instalación y configuración de Zoiper

Ahora instalaremos Zoiper en nuestros clientes para poder tener acceso al servicio de voz ip que nos brinda Asterisk.

Windows

Accedemos a la pagina oficial de Zoiper, y navegamos al apartado de descargas



Seleccionamos el cliente de windows y descargamos la versión gratuita

Zoiper 5

free VoIP softphone for non-commercial use

Desktop

 Windows	Download
 Mac	Download
 Linux	Download

Mobile

 Android	Download
iOS iOS	Download

Free €0	Premium €59.95	Custom On demand
		
Download	Purchase	Contact us
Basic functionality for occasional use non-commercial use only • Basic Functionality	Business functionality for the power user • Auto Provisioning • Contacts integration (Outlook, LDAP, Thunderbird, Mac OS X, Windows) • Browser integration (click to dial, open url) • Better quality/Lower bandwidth (G.729/H.264 codecs)	Fully customized solutions according to your specs • Your design • Your feature set • Locked to your service • Branded with your name and logo • Balance display

Instalamos Zoiper desde el ejecutable que acabamos de descargar

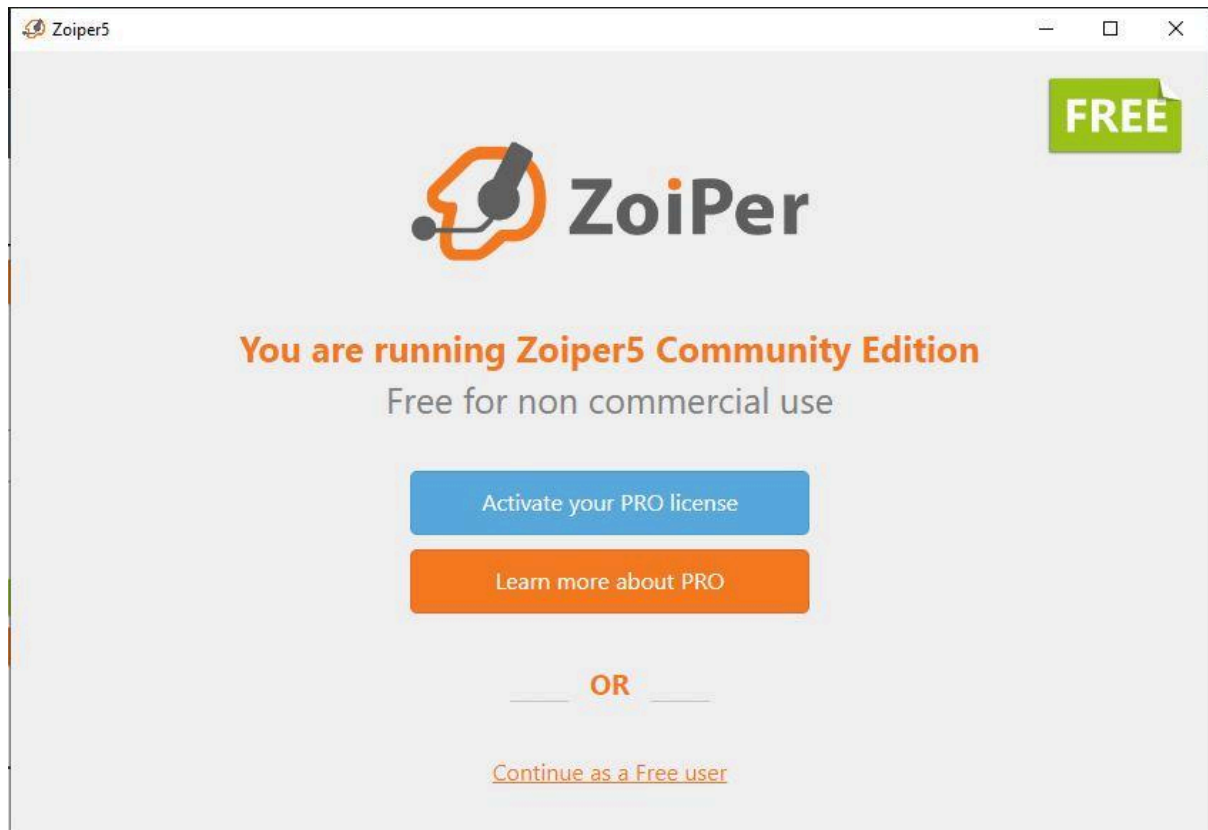


Seleccionamos siguiente y configuramos los parametros de instalación, como la ubicación de la misma.

Una vez finalizada la instalación, cerramos el asistente de instalación



Iniciamos Zoiper y continuamos como usuario gratuito



En los credenciales ponemos nuestro nombre de usuario o de extension en sip.conf seguido de @<ip_server_asterisk>, y la contraseña del usuario en cuestion

Zoiper5



The image shows the Zoiper login interface. On the left is a light gray sidebar with the Zoiper logo at the top. Below the logo are two input fields: the first contains 'ext1@192.168.1.189' and the second contains a masked password '....'. Below the password field is the text 'For example K23XWkd23'. At the bottom of the sidebar are two buttons: a green 'Login' button and an orange 'Create account' button. The main area on the right has a dark background with a white line-art illustration of a smiling face wearing a headset. Below the illustration, the text reads: 'Username and password', 'If you already have a VoIP account, please enter your login credentials to the left.', 'Most often these are provided either by your VoIP provider, or by your system administrator for your office's IP PBX.', and a small asterisked note: '*we will try to automatically find the necessary configuration for your account. If such does not exist in our database, you'll be asked to enter specific details.'

Zoiper

ext1@192.168.1.189

....

For example K23XWkd23

Login

Create account

Username and password

If you already have a VoIP account, please enter your login credentials to the left.

Most often these are provided either by your VoIP provider, or by your system administrator for your office's IP PBX.

*we will try to automatically find the necessary configuration for your account. If such does not exist in our database, you'll be asked to enter specific details.

Añadimos la IP de nuestro servidor en el apartado de proveedor. Si hubiesemos configurado otro puerto para el servicio tambien lo indicariamos en este apartado

Zoiper5



The image shows the Zoiper hostname selection screen. It has a light gray background. At the top, the text says 'Fill in your hostname and select your provider from the list'. Below this is a text input field containing '192.168.1.189'. Under the input field, there is explanatory text: 'This could also be called 'Domain', 'SIP server', 'Registrar' or 'SIP Proxy'. For example 'sip.example.com' or '123.21.123.32:5060''. At the bottom, there are two buttons: a white 'Back' button and a green 'Next' button.

Fill in your hostname and select your provider from the list

192.168.1.189

This could also be called 'Domain', 'SIP server', 'Registrar' or 'SIP Proxy'. For example 'sip.example.com' or '123.21.123.32:5060'

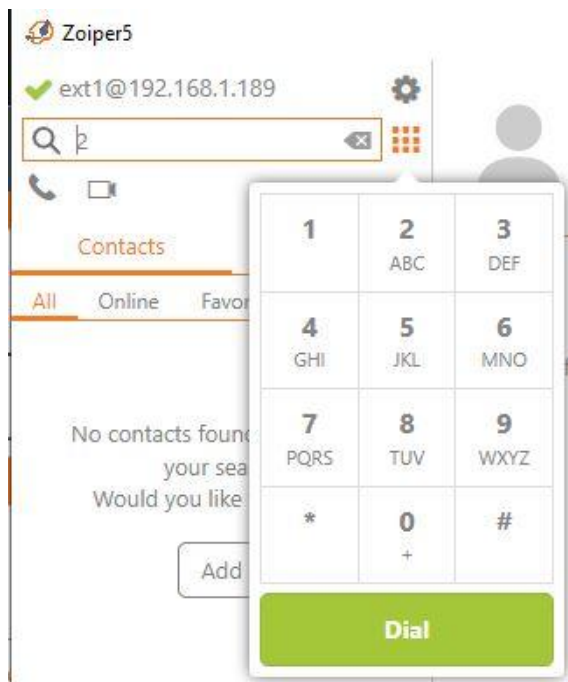
Back

Next

Acto seguido Zoiper buscara las posibles configuraciones. Obtenemos SIP UDP ya que es la que hemos configurado

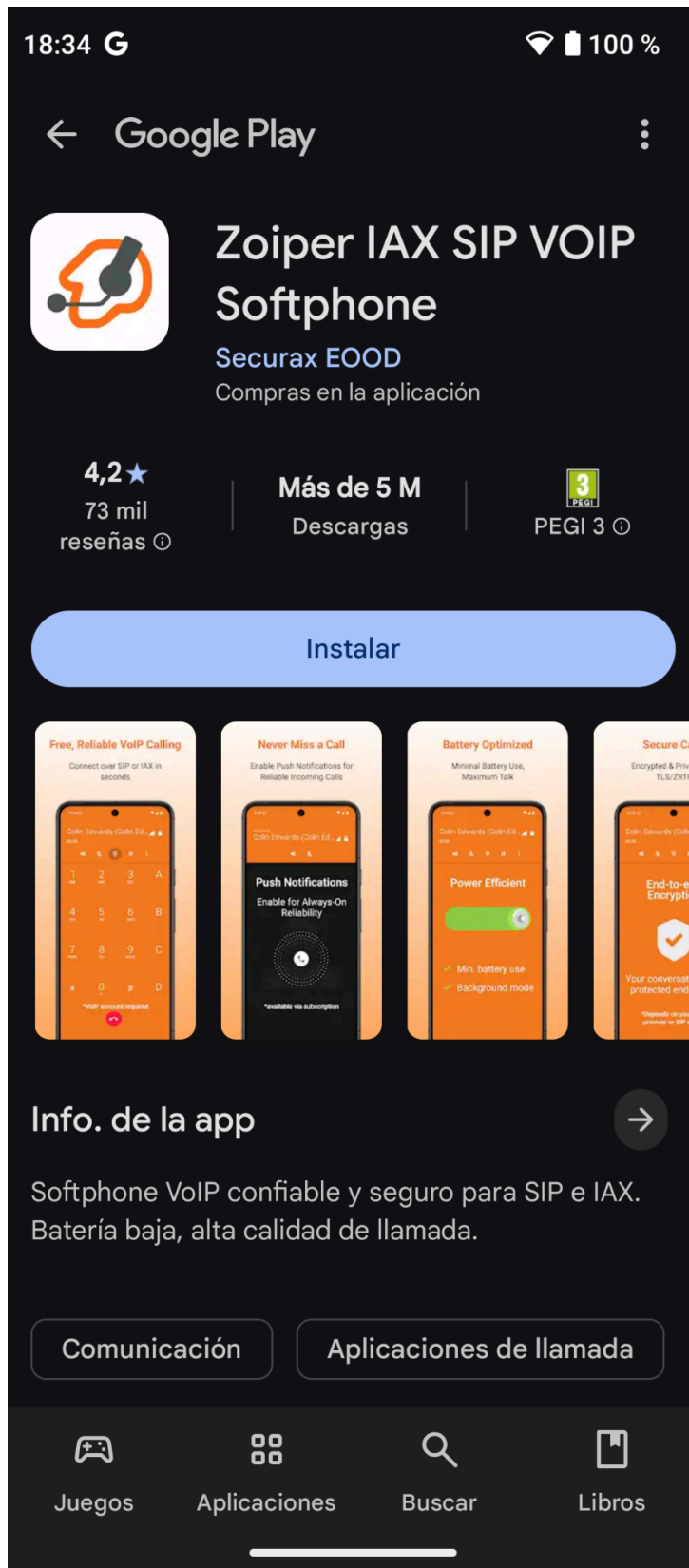


Y finalmente tendremos acceso a nuestro usuario



Android

Descargamos e instalamos la aplicación oficial desde la Play Store



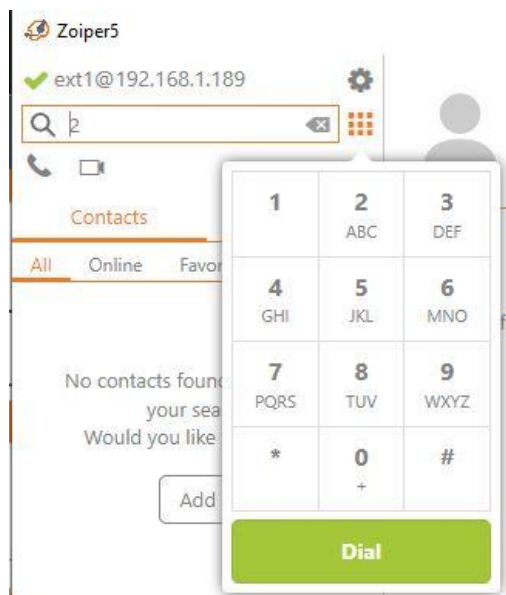
Accederemos al mismo menu de configuración que en windows y seguimos los mismos pasos

The first screenshot shows the 'Configuracion de cuenta' screen with the Zoiper logo and fields for 'Nombre de usuario @ PBX/proveedor de VoIP' (filled with 'ext2@192.168.1.189') and 'Contraseña'. Below are 'Login' and 'List de proveedores' buttons, and a QR code with the text 'Iniciar sesión con...'. The second screenshot shows the same screen with the 'nombre de server o proveedor' field filled with '192.168.1.189' and a 'Siguiete' button. The third screenshot shows the 'Seleccione una de las siguientes configuraciones' screen with four options: SIP TLS (No encontrado), SIP TCP (No encontrado), SIP UDP (Encontrado), and IAX UDP (No encontrado). A 'Terminar' button is at the bottom right.

Una vez finalizado el proceso tendremos acceso al menu principal

Comprobación

Realizaremos una llamada de prueba entre nuestros clientes para comprobar que funcionan correctamente y nuestro servicio de voz ip funciona. Marcamos a nuestro cliente android desde windows



Automaticamente recibiremos la llamada en nuestro cliente android. Aceptamos la llamada y comprobamos que efectivamente funciona correctamente

